

A Literature-Implied Negative Answer to the Connes–Kirchberg Problem in arXiv:1101.4195

FA/Banach run packet

2026-06-14

Status

This is a `literature_implied_answers` packet. It records that the Connes–Kirchberg/QWEP problem stated in Pisier’s survey [1] has a negative answer as a consequence of the later $\text{MIP}^* = \text{RE}$ theorem of Ji–Natarajan–Vidick–Wright–Yuen [2], together with Kirchberg’s equivalence between the tensor-norm/QWEP formulation and Connes’ embedding problem as stated in the source paper. This is not an original proof from the run.

Original Problem Evidence

The relevant questions appear as Problems 12.9 and 12.10 on page 41 of the downloaded source PDF. In the source TeX they are labelled `prbl10.4` and `prbl10.4bis`.

soon after, several different, more efficient proofs were found (see chapters 21 and 22 in [129]).

The next problem is currently perhaps the most important open question in Operator Algebra Theory. Indeed, as Kirchberg proved (see the end of [84]), this is also equivalent to a fundamental question raised by Alain Connes [26] on von Neumann algebras: whether any separable II_1 -factor (or more generally any separable finite von Neumann algebra) embeds into an ultraproduct of matrix algebras. We refer the reader to [110] for an excellent exposition of this topic.

Problem 12.9. Let $A = C^*(\mathbb{F}_N)$ ($N > 1$). Is there a unique C^* -norm on $A \otimes A$? Equivalently, is (A, A) a nuclear pair?

Equivalently:

Problem 12.10. Is every C^* -algebra QWEP ?

Curiously, there seems to be a connection between GT and Problem 12.9. Indeed, like GT, the latter problem can be phrased as the identity of two tensor products on $\ell_1 \otimes \ell_1$ or more precisely on

In words, the source asks whether, for $A = C^*(\mathbb{F}_N)$, $N > 1$, there is a unique C^* -norm on $A \otimes A$, equivalently whether (A, A) is a nuclear pair. The adjacent formulation asks whether every C^* -algebra is QWEP. The paragraph immediately above the problem states that Kirchberg proved equivalence with Connes’ embedding problem for finite von Neumann algebras.

Supporting Answer Evidence

The paper $\text{MIP}^* = \text{RE}$ states in its abstract that, using the known connection through Tsirelson’s problem, the results refute Connes’ embedding conjecture.

MIP* = RE

Zhengfeng Ji^{*1}, Anand Natarajan^{†2,3}, Thomas Vidick^{‡3}, John Wright^{§2,3,4}, and Henry Yuen^{¶5}

¹*Centre for Quantum Software and Information, University of Technology Sydney*

²*Institute for Quantum Information and Matter, California Institute of Technology*

³*Department of Computing and Mathematical Sciences, California Institute of Technology*

⁴*Department of Computer Science, University of Texas at Austin*

⁵*Department of Computer Science and Department of Mathematics, University of Toronto*

November 7, 2022

Abstract

We show that the class MIP* of languages that can be decided by a classical verifier interacting with multiple all-powerful quantum provers sharing entanglement is equal to the class RE of recursively enumerable languages. Our proof builds upon the quantum low-degree test of (Natarajan and Vidick, FOCS 2018) and the classical low-individual degree test of (Ji, et al., 2020) by integrating recent developments from (Natarajan and Wright, FOCS 2019) and combining them with the recursive compression framework of (Fitzsimons et al., STOC 2019).

An immediate byproduct of our result is that there is an efficient reduction from the Halting Problem to the problem of deciding whether a two-player nonlocal game has entangled value 1 or at most $\frac{1}{2}$. Using a known connection, undecidability of the entangled value implies a negative answer to Tsirelson's problem: we show, by providing an explicit example, that the closure C_{qa} of the set of quantum tensor product correlations is strictly included in the set C_{qc} of quantum commuting correlations. Following work of (Fritz, Rev. Math. Phys. 2012) and (Junge et al., J. Math. Phys. 2011) our results provide a refutation of Connes' embedding conjecture from the theory of von Neumann algebras.

Goldbring's later guided tour records the negative solution to CEP and describes the traditional route through Kirchberg's QWEP problem.

THE CONNES EMBEDDING PROBLEM: A GUIDED TOUR

ISAAC GOLDBRING

ABSTRACT. The Connes Embedding Problem (CEP) is a problem in the theory of tracial von Neumann algebras and asks whether or not every tracial von Neumann algebra embeds into an ultrapower of the hyperfinite II_1 factor. The CEP has had interactions with a wide variety of areas of mathematics, including C^* -algebra theory, geometric group theory, free probability, and noncommutative real algebraic geometry (to name a few). After remaining open for over 40 years, a negative solution was recently obtained as a corollary of a landmark result in quantum complexity theory known as $\text{MIP}^* = \text{RE}$. In these notes, we introduce all of the background material necessary to understand the proof of the negative solution of the CEP from $\text{MIP}^* = \text{RE}$. In fact, we outline two such proofs, one following the “traditional” route that goes via Kirchberg’s QWEP problem in C^* -algebra theory and Tsirelson’s problem in quantum information theory and a second that uses basic ideas from logic.

Idea of the Proof

The source problem is an equivalent formulation of CEP: a positive answer to the tensor-norm question for full group C^* -algebras of nonabelian free groups is the same as a positive solution of the Kirchberg/QWEP/Connes embedding problem. The later $\text{MIP}^* = \text{RE}$ theorem proves, through the quantum-correlation formulation, that CEP is false. Transporting that negative result back through Kirchberg’s equivalence gives the negative answer to the tensor-norm and QWEP formulations in Pisier’s source paper.

Literature-Implied Answer. *Problems 12.9 and 12.10 of [1] have negative answers. In the Kirchberg formulation, the full group C^* -algebra of a nonabelian free group does not have a unique C^* -norm on its algebraic tensor square, and not every C^* -algebra is QWEP.*

Proof. Pisier’s source states Problem 12.9 as the question whether, for $A = C^*(\mathbb{F}_N)$, $N > 1$, there is a unique C^* -norm on $A \otimes A$, equivalently whether (A, A) is a nuclear pair. It then states the equivalent QWEP formulation, Problem 12.10, asking whether every C^* -algebra is QWEP. The surrounding discussion records Kirchberg’s equivalence between this problem and Connes’ embedding problem.

Ji–Natarajan–Vidick–Wright–Yuen prove $\text{MIP}^* = \text{RE}$ and state that, following the known connections through Tsirelson’s problem, their result gives a refutation of Connes’ embedding conjecture [2]. Goldbring’s guided tour gives a post- $\text{MIP}^* = \text{RE}$ account of the same implication and explicitly describes the traditional route through Kirchberg’s QWEP problem [3].

Therefore Connes’ embedding problem is false. Since Pisier’s tensor-norm and QWEP questions are equivalent to it, those questions are false as well. Thus the algebraic tensor product $C^*(\mathbb{F}_N) \otimes C^*(\mathbb{F}_N)$ in Kirchberg’s nonabelian free-group formulation cannot have a unique C^* -norm, and there must exist C^* -algebras that are not QWEP. $\square$

Limitations

This packet addresses only the Connes–Kirchberg/QWEP problem cluster in Pisier’s survey. The survey contains many other open problems, including questions about exact Grothendieck constants, noncommutative Khintchine inequalities, and multilinear operator-space Grothendieck-type problems. Those are not claimed here.

Verification Notes

The source and supporting papers are distinct. The route is an implication through Kirchberg’s equivalence, so the packet belongs in the literature-implied bucket rather than the already-answered bucket. The local duplicate search covered the arXiv ids 1101.4195, 2001.04383, and 2109.12682, together with the phrases Connes Embedding, QWEP, Kirchberg, and MIP*. No existing packet or attempt covered this exact status identification. No computation is involved.

Human Review Recommendation

Check three points: first, that Pisier’s Problem 12.9/12.10 is indeed the Kirchberg/QWEP formulation equivalent to CEP; second, that [2] establishes the negative CEP consequence; and third, that the nonabelian free-group tensor-norm formulation in the source is covered by the standard Kirchberg equivalence. If these are accepted, the negative answer follows.

References

- [1] G. Pisier, *Grothendieck’s theorem, past and present*, Bull. Amer. Math. Soc. 49 (2012), no. 2, 237–323; arXiv:1101.4195.
- [2] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $\text{MIP}^* = \text{RE}$, Commun. ACM 64 (2021), no. 11, 131–138; arXiv:2001.04383.
- [3] I. Goldbring, *The Connes Embedding Problem: A guided tour*, arXiv:2109.12682.